

Fundamentals of Computer Programming (I)

Homework - Lecture 2

3. Among the following identifiers, which are legal C variable names?

a_qwe

zx-123

\$a234

_a_sdf

qw.c

x/y

a*b

%jkh

xy%c

_1234

1234_

12_34

c1_2

x4_5_6

new_r

root_1

_root1

12345

4. In a 32-bit compiler environment, given the following declaration statements,

```
short x = 0xffff;
```

```
long y = 0xffffL;
```

What are the actual decimal values assigned to variables x and y after executing the following code?

➤ Computer-based experiments

(1) Write a program to print and view the double-precision floating-point values of 97.6875 and -97.6875. Display their IEEE 754 storage format in hexadecimal by bytes to verify if the results are consistent with those given in the textbook. Also, print and view the single-precision floating-point values (float) of 97.6875 and -97.6875 in the computer, showing their IEEE 754 storage format in hexadecimal by bytes.

(2) Write a program to print the hexadecimal representation by bytes of the storage of 100 and -100 in the computer for double, float, long, int, short, and char data types respectively. Analyze why the results are different for each data type.

➤ Attached: An example program that prints the hexadecimal representation of the internal values of double and float type variables:

```
#include <stdio.h>

int main() {
    double x = 97.6875;
    float y = 97.6875;
    unsigned char *p;
    p = (unsigned char *)&x;
    printf("%02X %02X %02X %02X %02X %02X %02X %02X\n",
           *p, *(p+1), *(p+2), *(p+3), *(p+4), *(p+5), *(p+6), *(p+7));
    p = (unsigned char *)&y;
    printf("%02X %02X %02X %02X\n", *p, *(p+1), *(p+2), *(p+3));
}
```

The program sequentially prints the hexadecimal value of each byte pointed to by the pointer.